

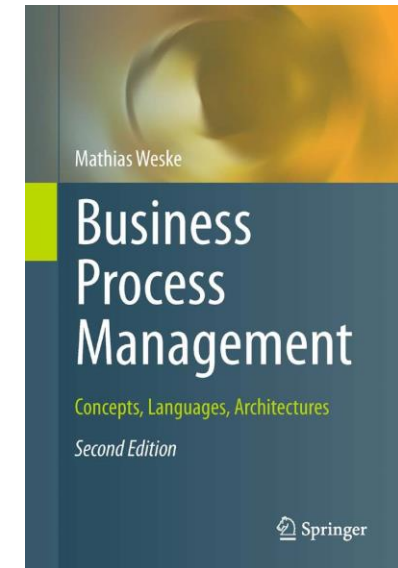
BUSINESS PROCESS MANAGEMENT

5 – Process Choreographies

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March 2025

Evolution of Enterprise System Architectures

- This PowerPoint file contains figures used in
 - Mathias Weske, Business Process Management: Concepts, Languages, Architectures 2012, XIV, 403 p. 300 illus., Hardcover ISBN: 978-3-642-28615-5, eBook ISBN: 978-3-642-28616-2 © Springer-Verlag Berlin Heidelberg 2012, 2007
- This module is based on that book



Process choreography & deadlock's

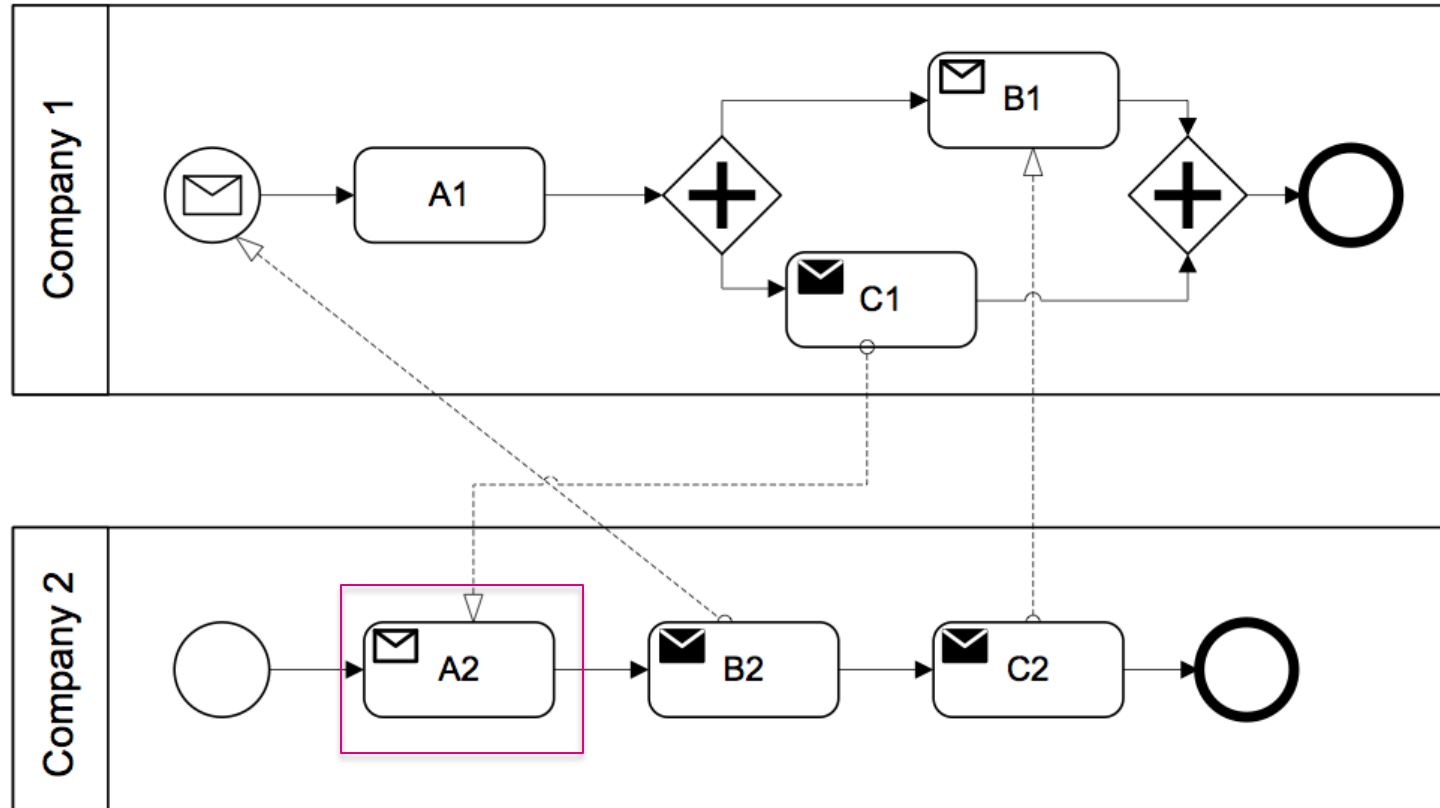
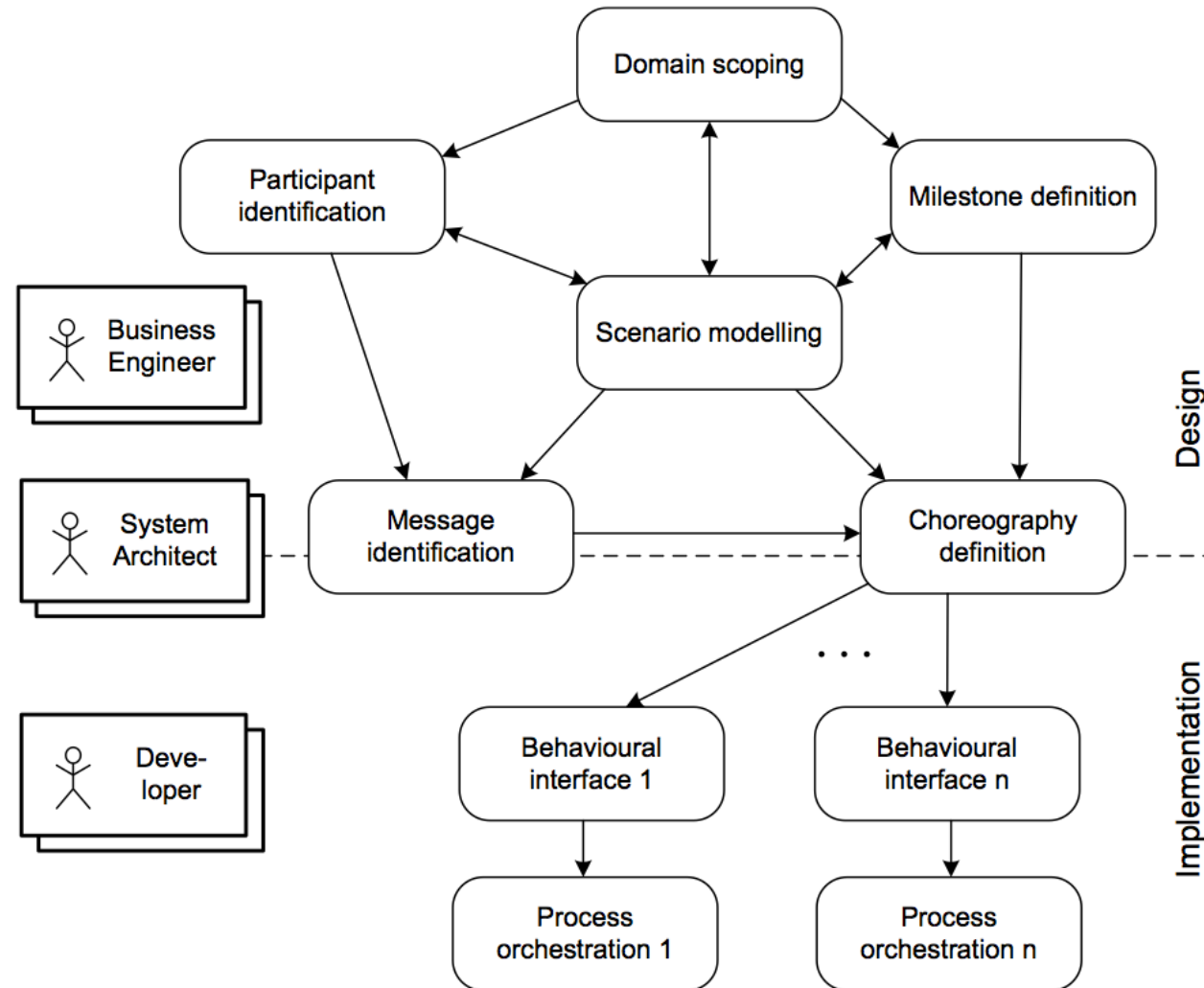


Fig. 5.1. Deadlock of interacting process orchestrations

Development of process choreographies



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Fig. 5.4. Phases during choreography design and implementation

Contents – Process Choreographies

- High-level Structure Design (Participant Identification Phase)
 - Identify participants roles and their communication structures
- High-level Behavioural Design (Milestone Definition Phase)
 - Specify milestones of collaboration
 - Specify order in which milestones are reached
- Collaboration Scenarios (Choreography Definition Phase)
 - Refine high-level choreographies
 - Introduce dedicated collaboration scenarios
 - Relate reaching of milestones to communication between process participants
- Behavioural interfaces (from Collaboration Scenarios)
 - For each participant role derive a behavioural interface

High-level structural model

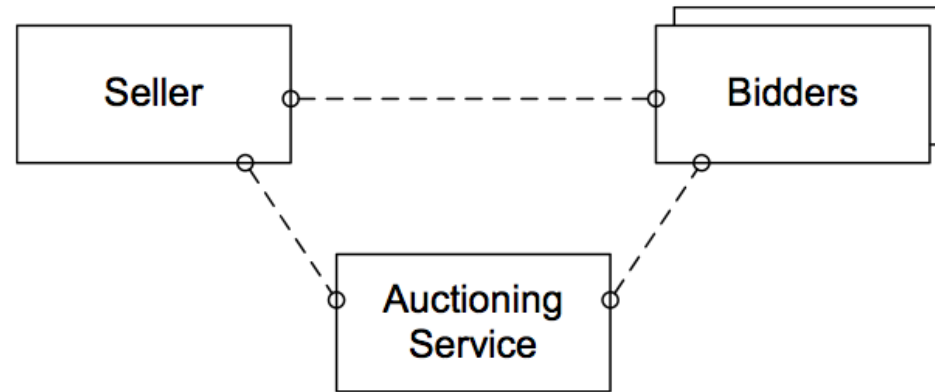


Fig. 5.5. High-level structural model of participants in bidding scenario

Milestones for high-level behavioural model (1/2)

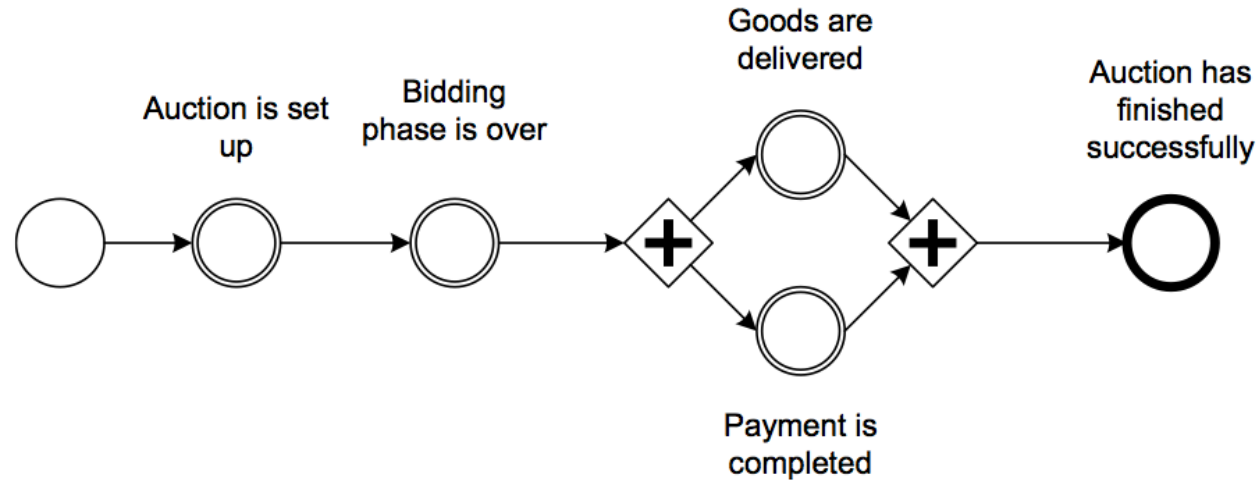


Fig. 5.6. High-level behavioural model for bidding scenario, represented by milestones

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- In figure 5.6, milestones are defined by circles, where the initial milestone has a single border, the intermediate milestones have double borders, and the ultimate goal milestone has a bold border

Milestones for high-level behavioural model (2/2)

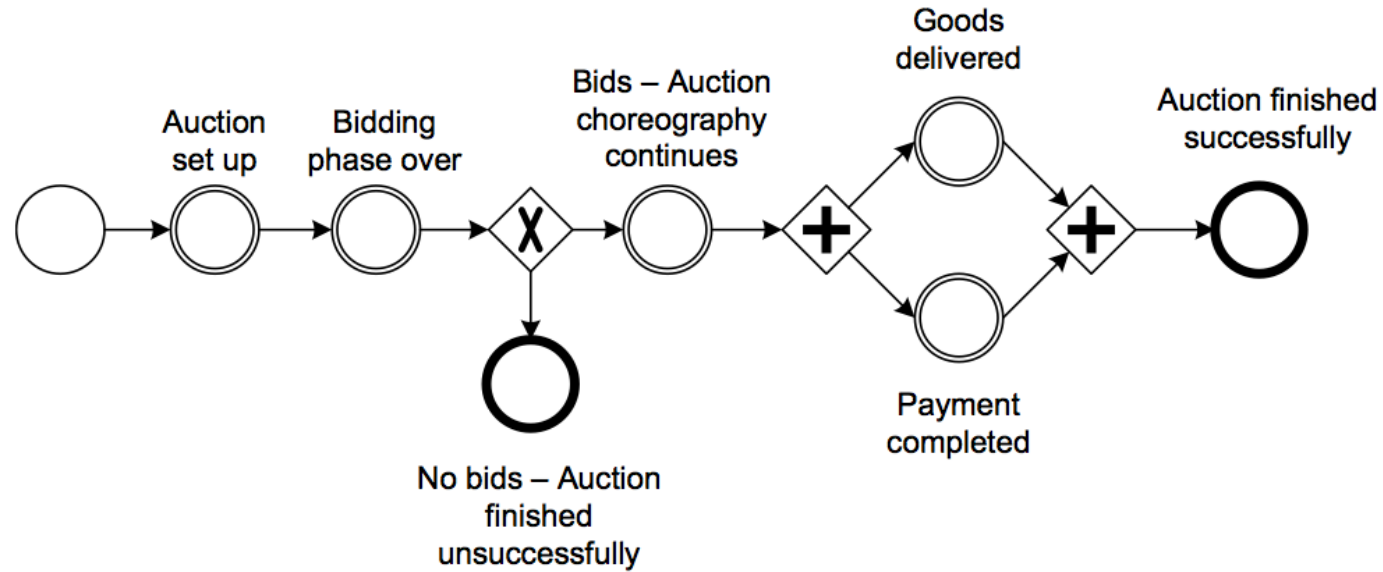
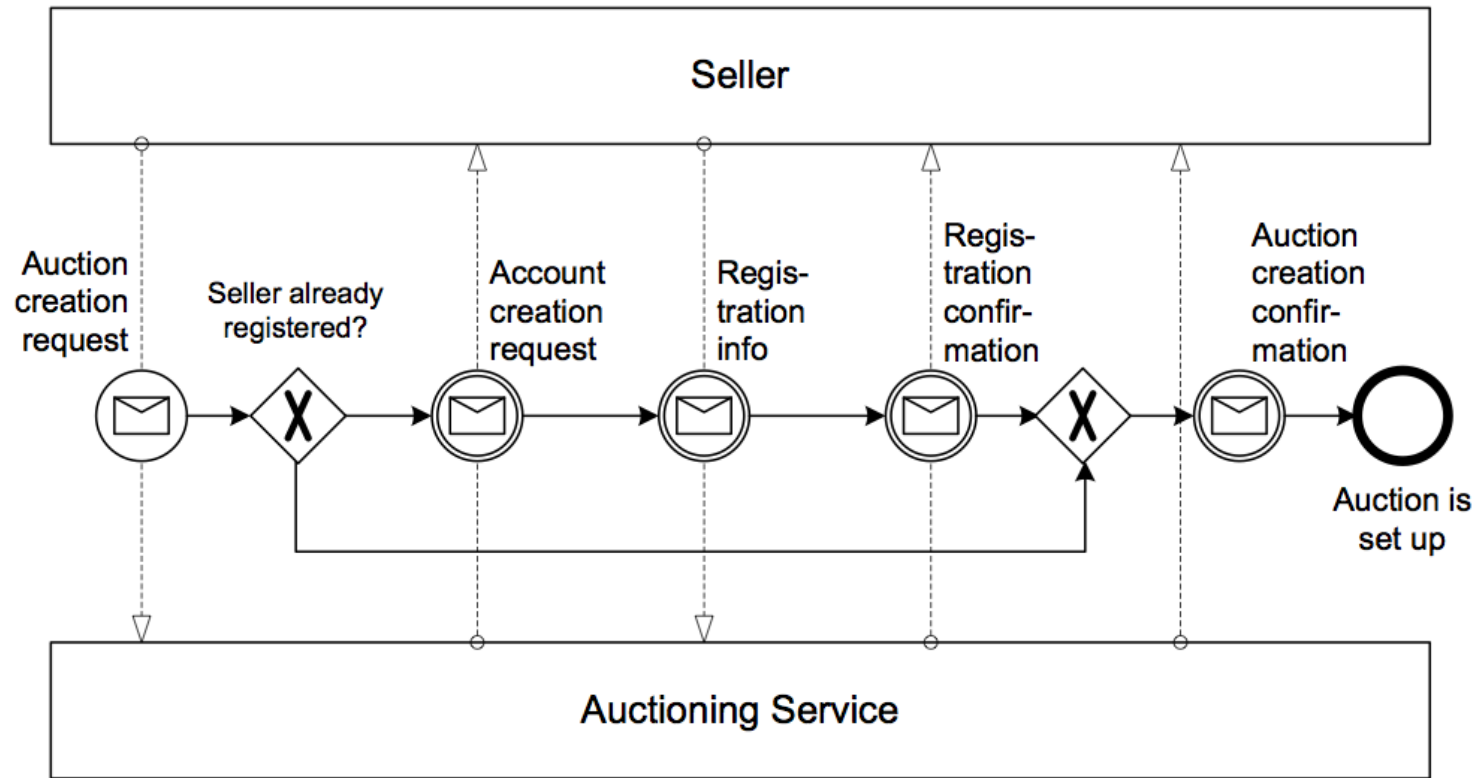


Fig. 5.7. High-level behavioural model for bidding scenario, with different outcomes

Collaboration scenario



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Fig. 5.8. Collaboration scenario: reaching milestones through interactions

Design of behavioural interface's

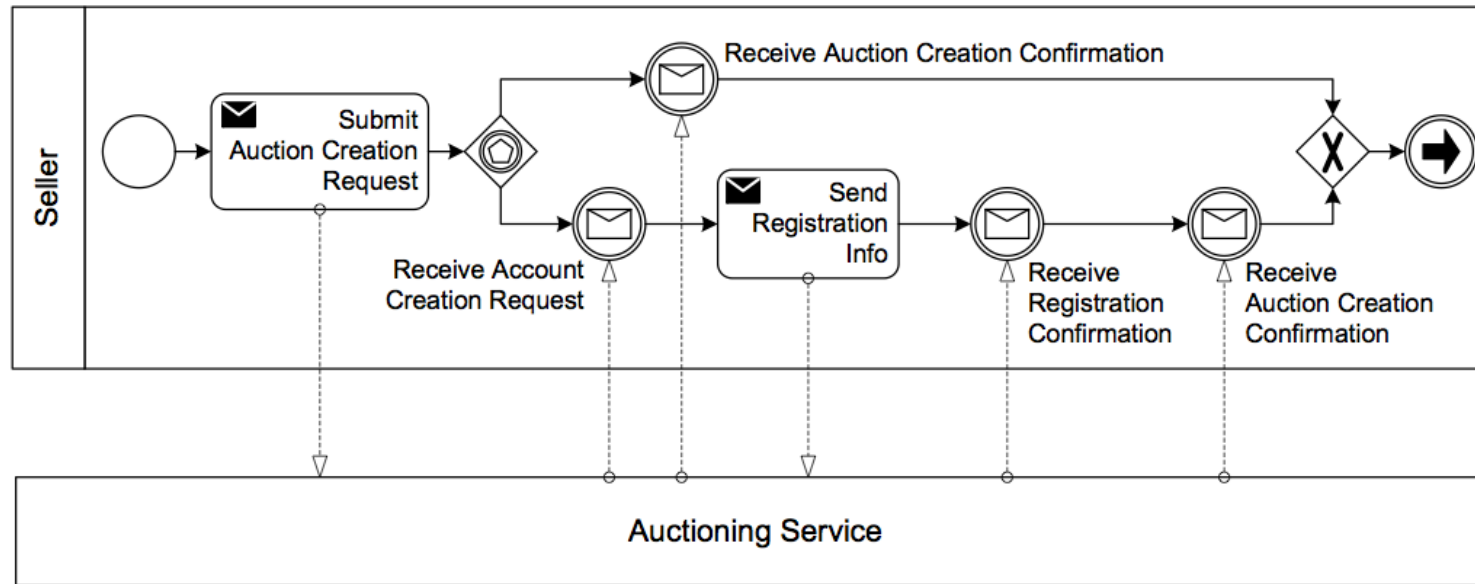
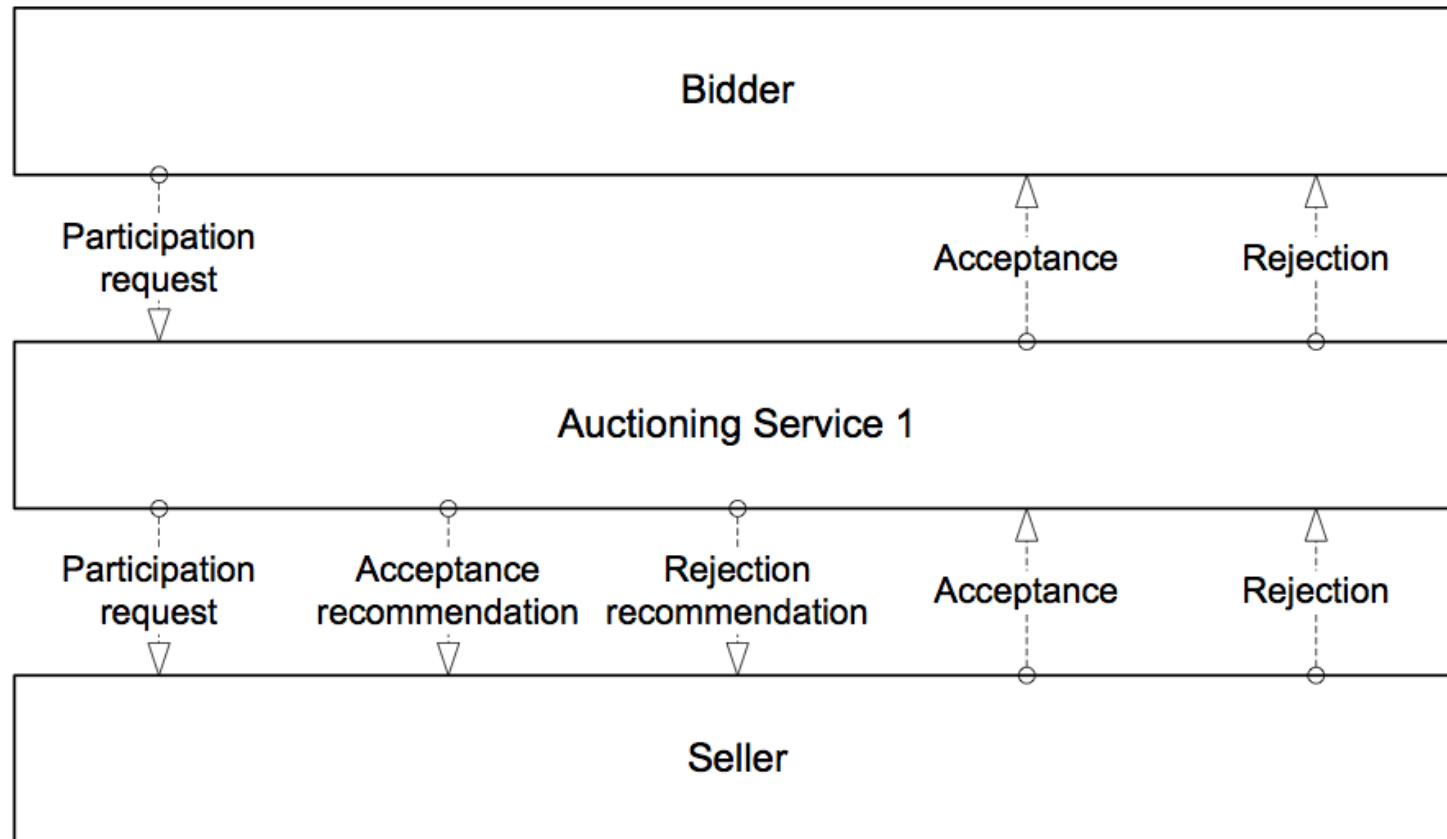


Fig. 5.9. Behavioural interface for seller

Interactions of participants in auctioning scenario



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Fig. 5.10. Interactions of participants in auctioning scenario

Service interaction patterns: send

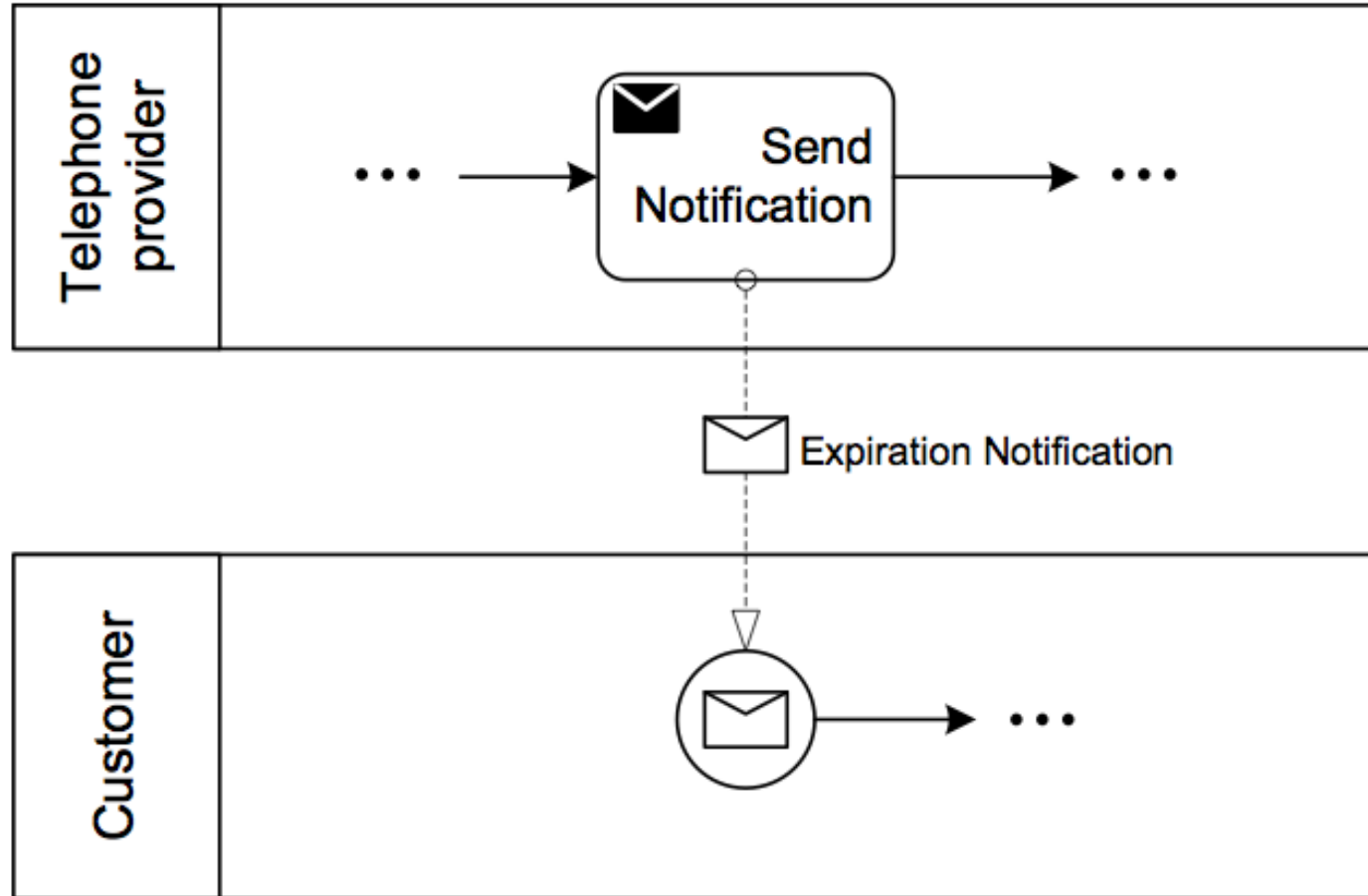


Fig. 5.22. Send pattern

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Service interaction patterns: receive

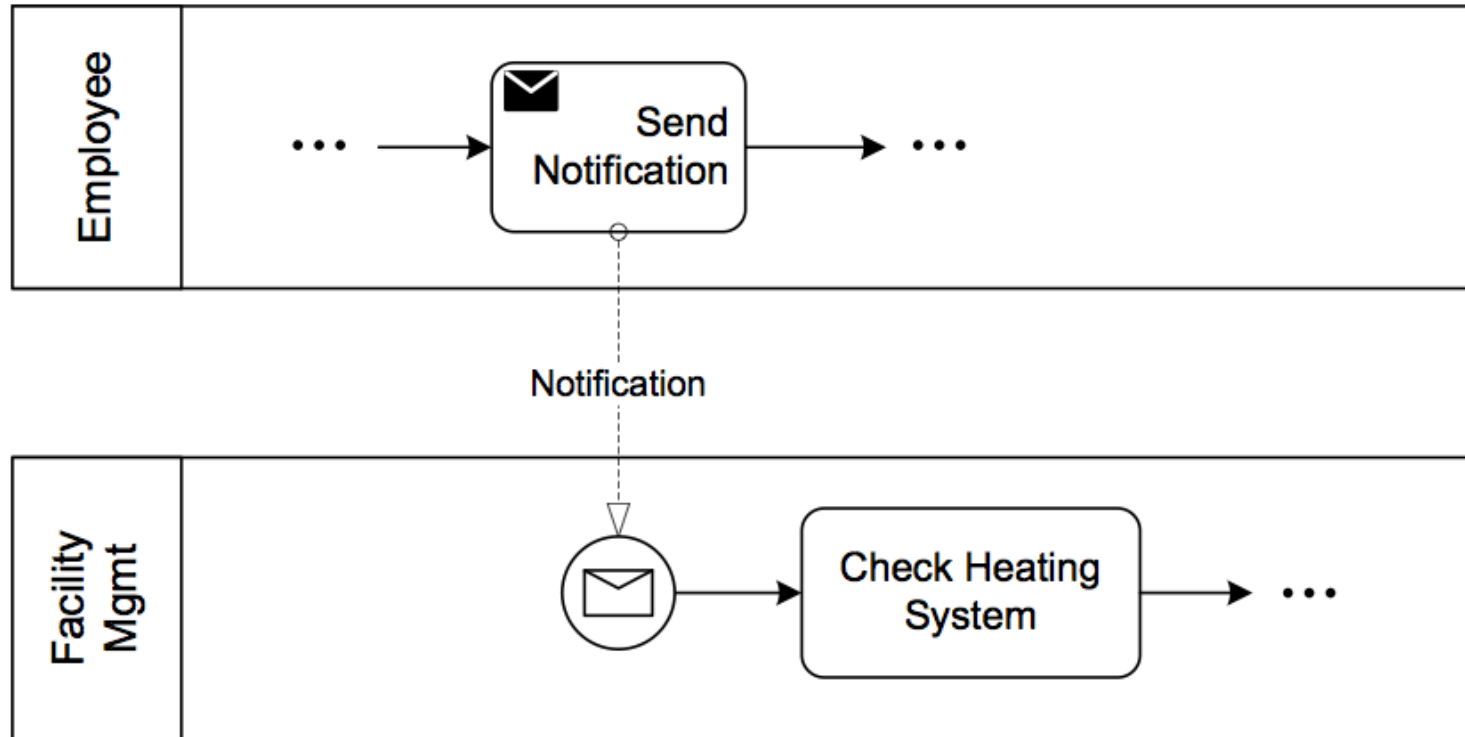


Fig. 5.23. Receive pattern

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Service interaction patterns: send & receive

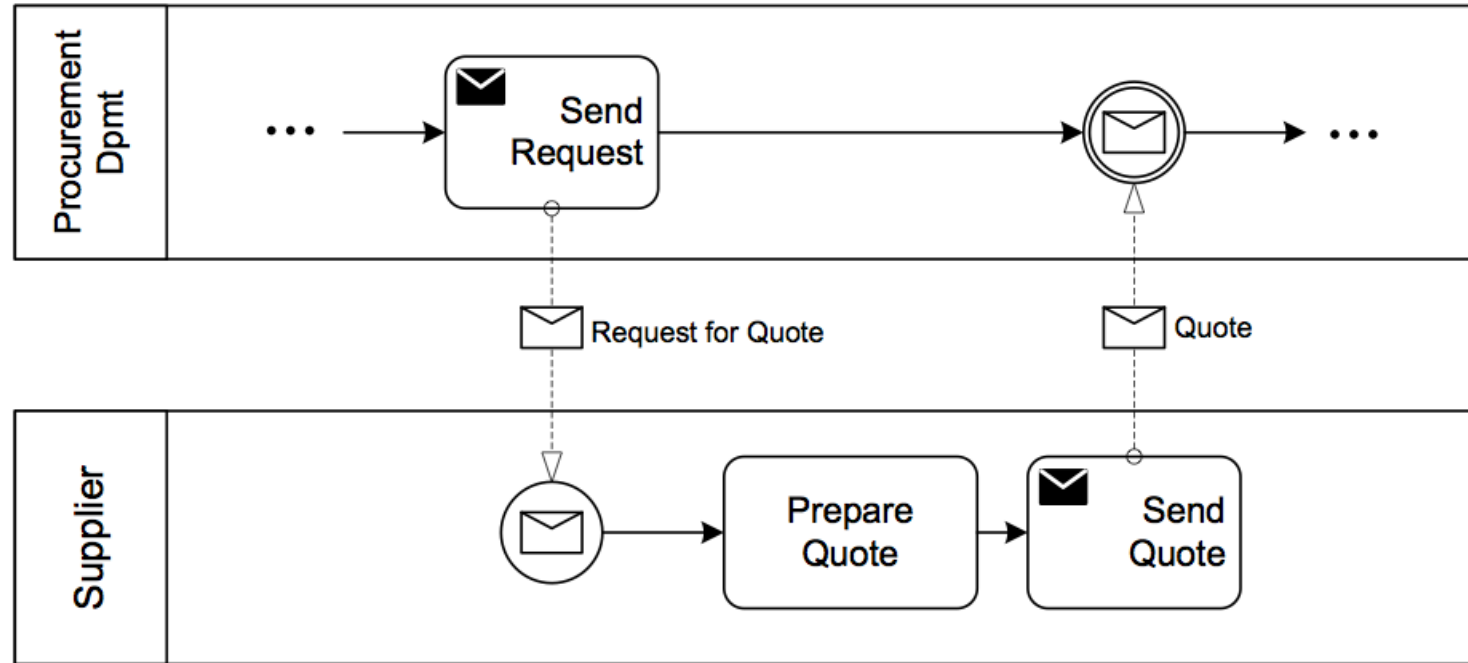
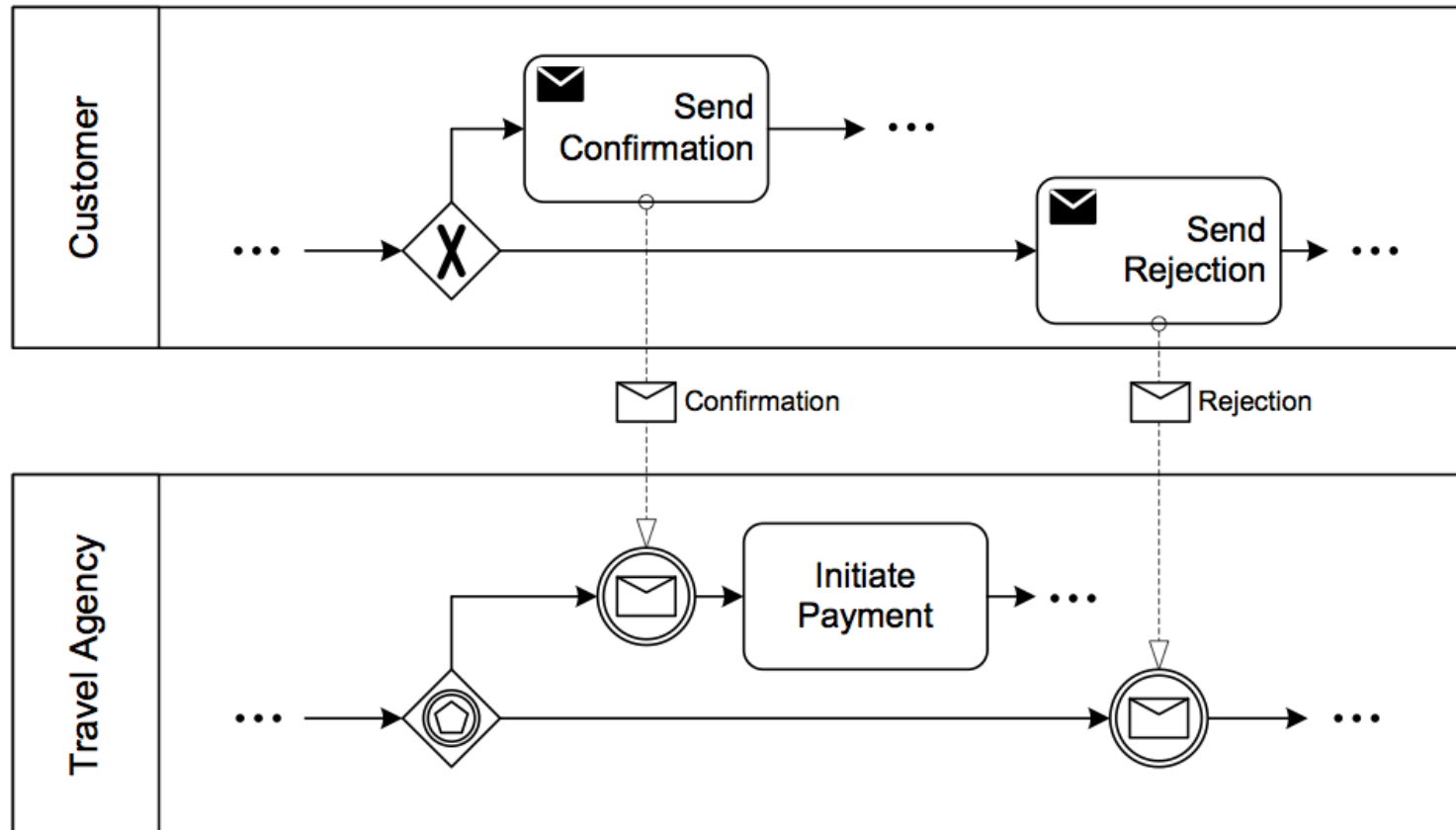


Fig. 5.24. Send/receive pattern

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Racing Incoming Messages



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Fig. 5.25. Racing incoming messages pattern

One-To-Many Send

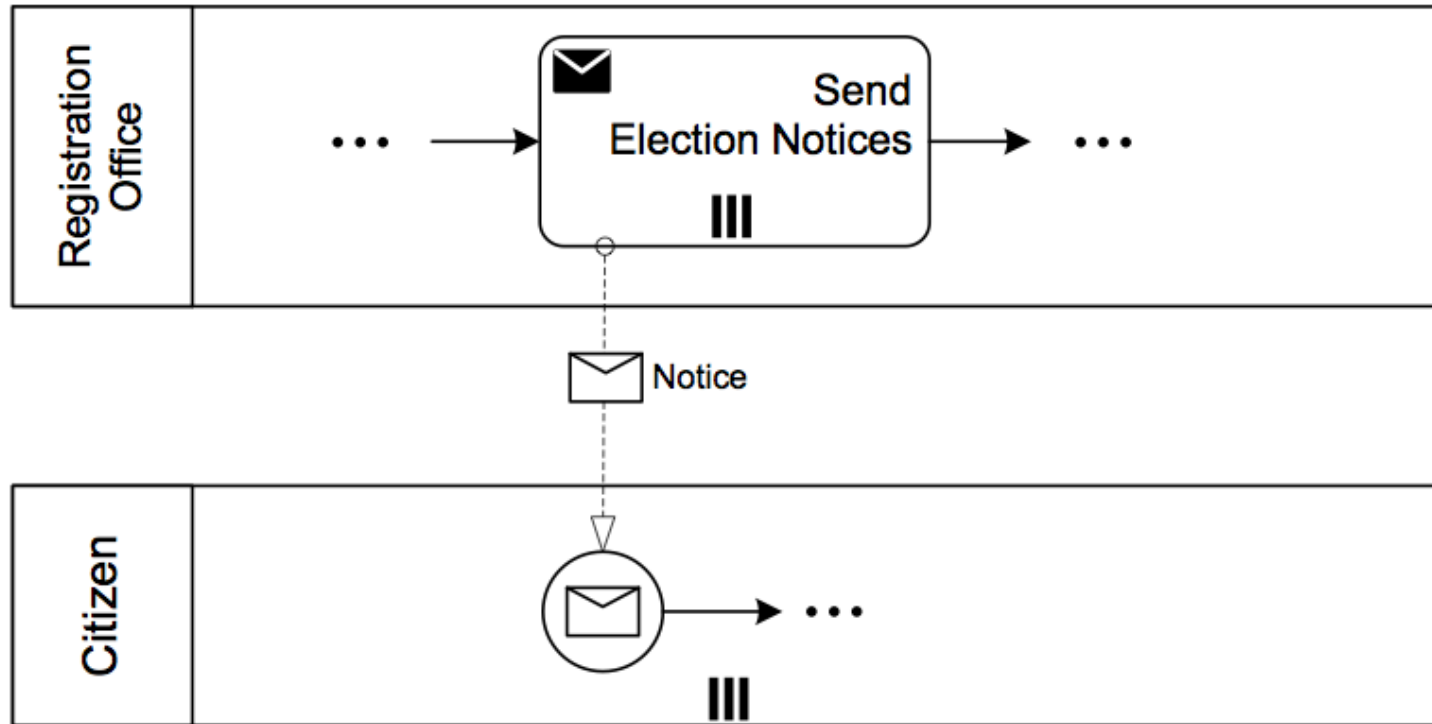
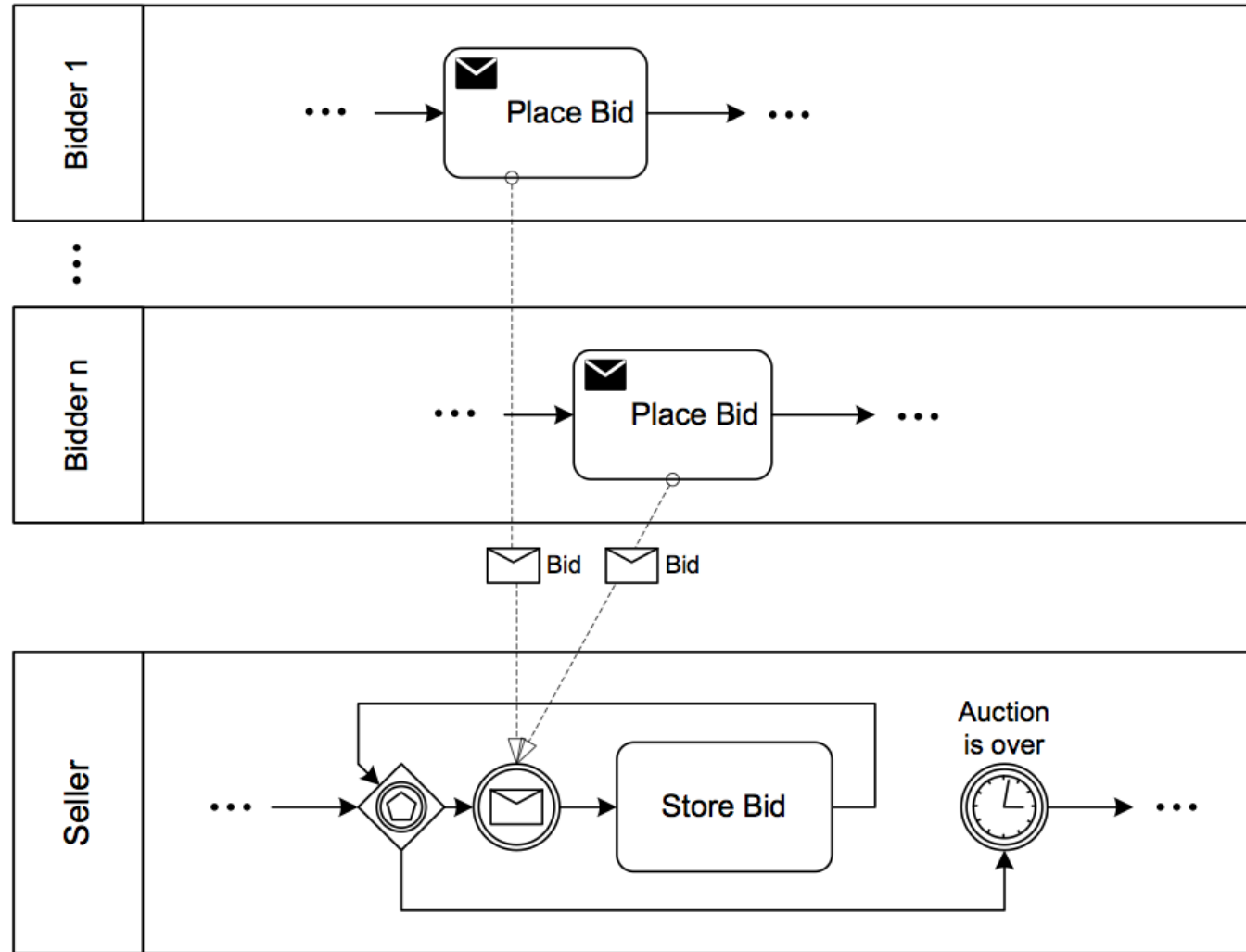


Fig. 5.26. One-to-many send pattern

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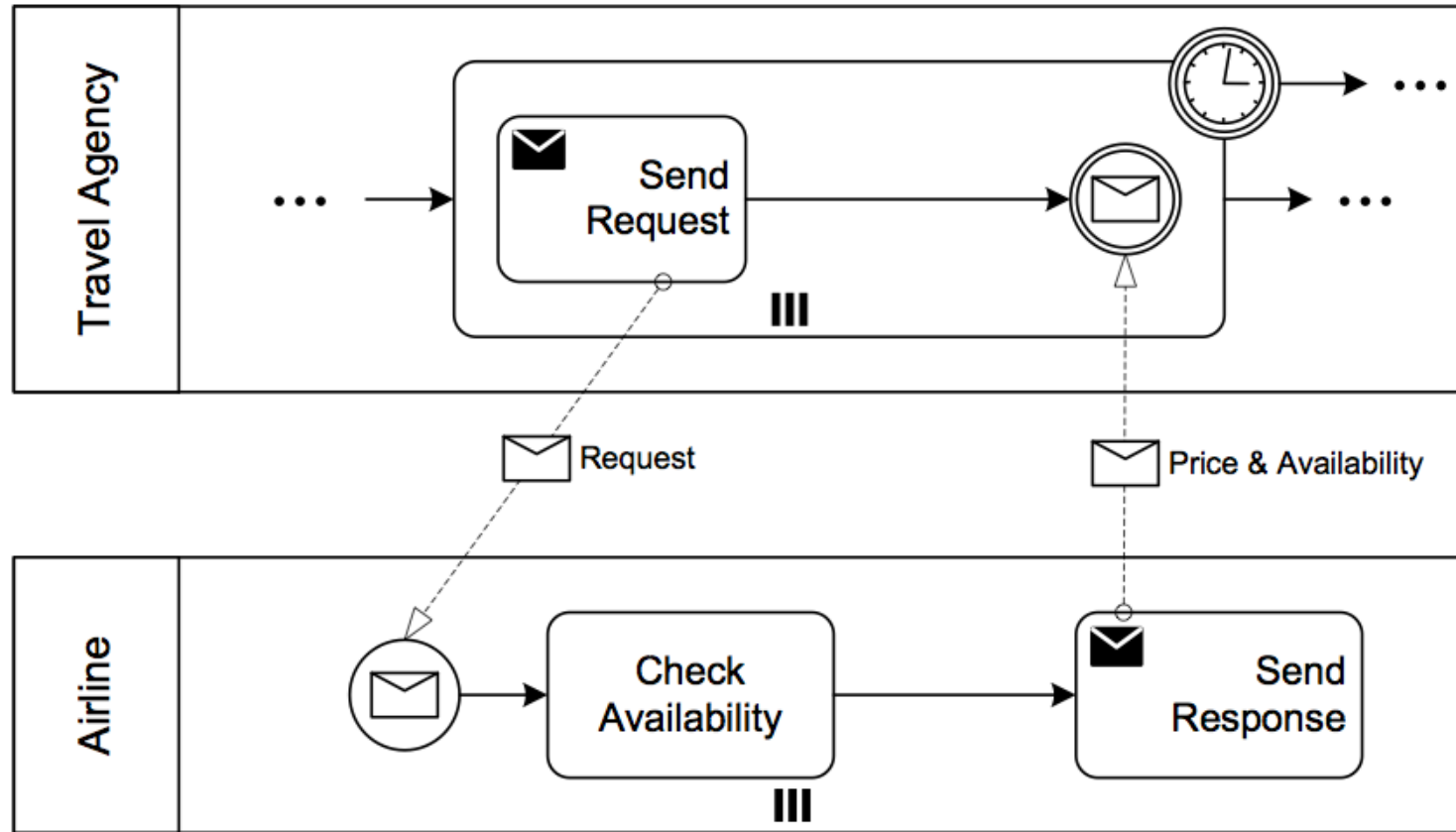
One-From-Many Receive



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Fig. 5.27. One-from-many receive pattern

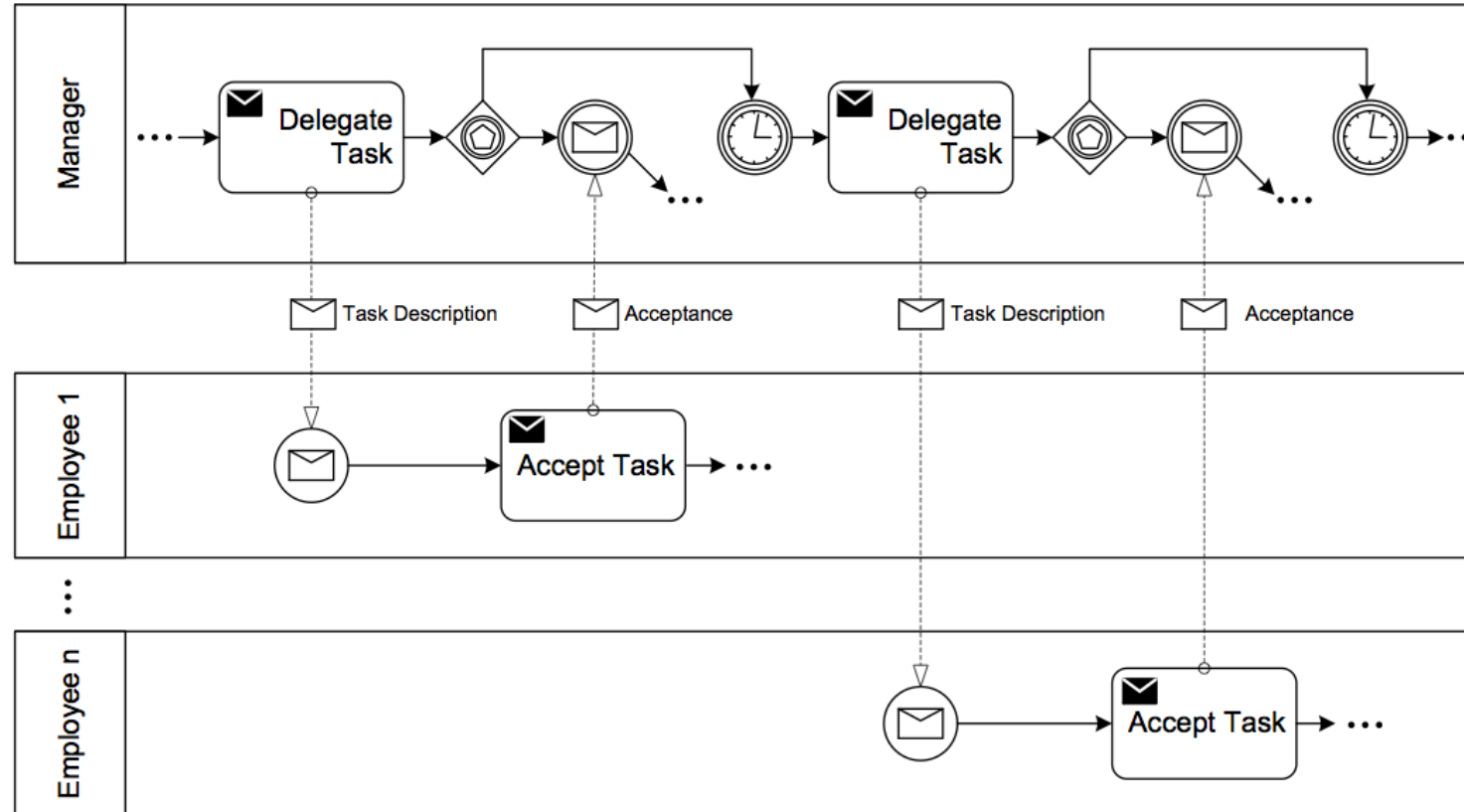
One-To-Many Send/Receive



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Fig. 5.28. One-to-many send/receive pattern

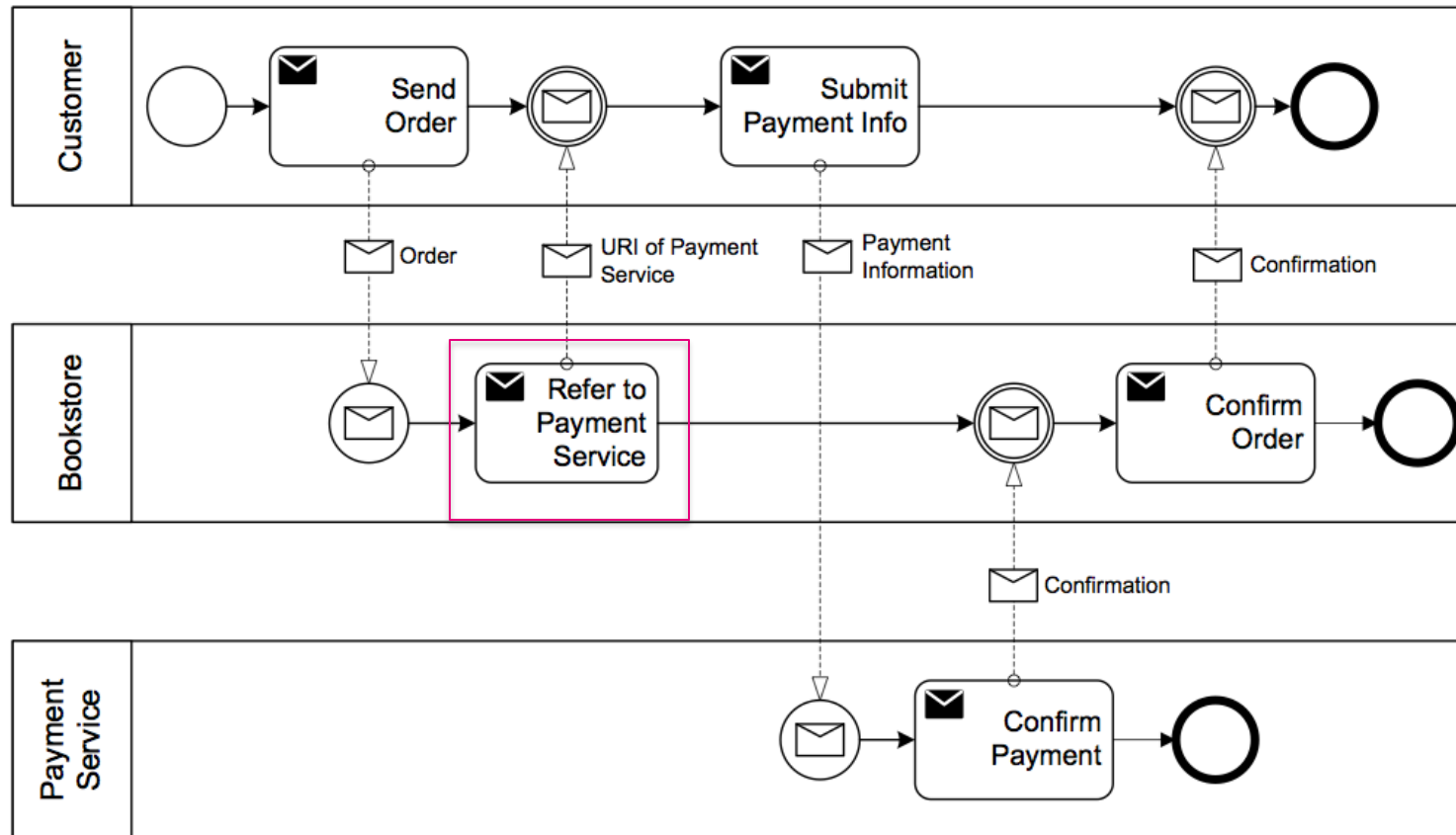
Contingent Requests



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Fig. 5.29. Contingent requests pattern

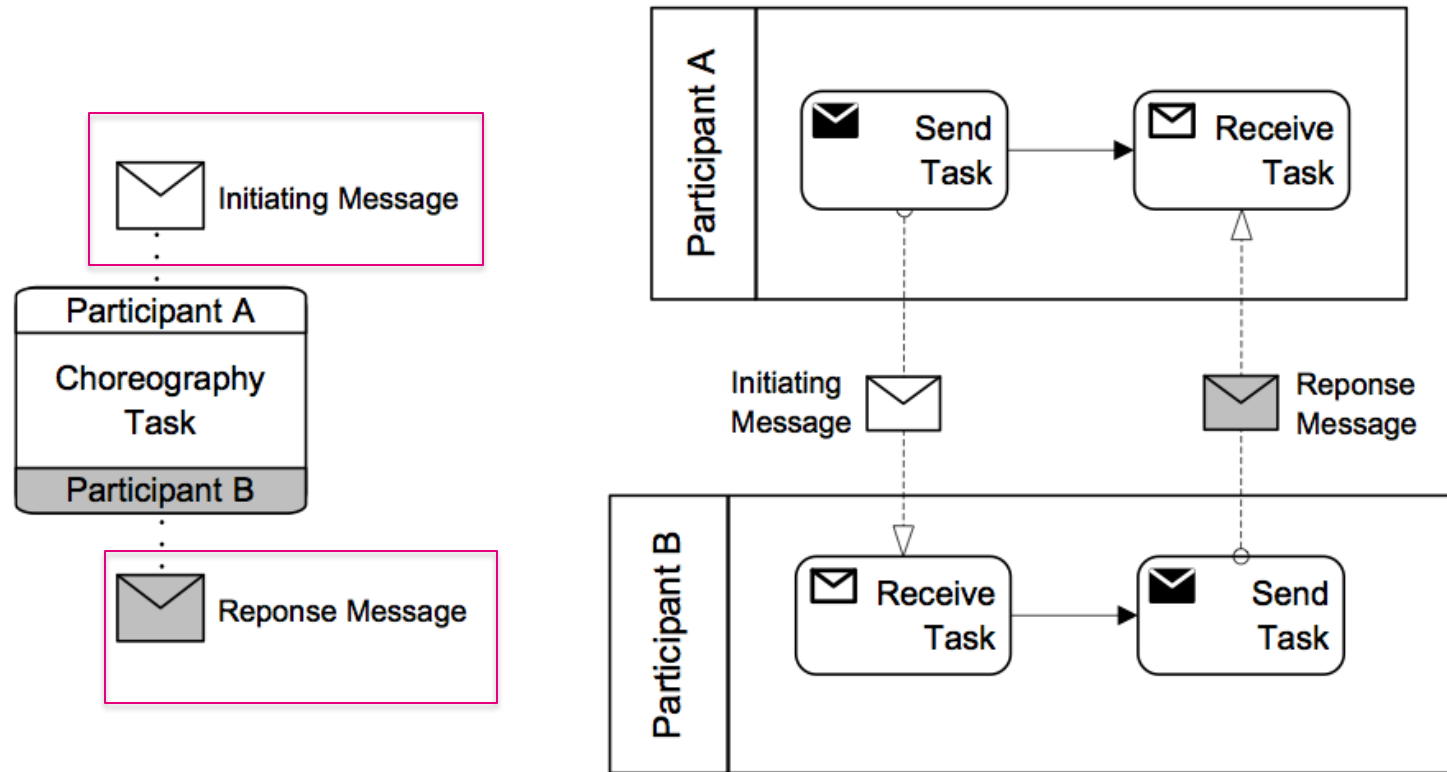
Request With Referral



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Fig. 5.30. Example involving request with referral pattern

Choreography Diagrams (1/2)



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Fig. 5.39. Choreography task and corresponding process diagram

Choreography Diagrams (2/2)

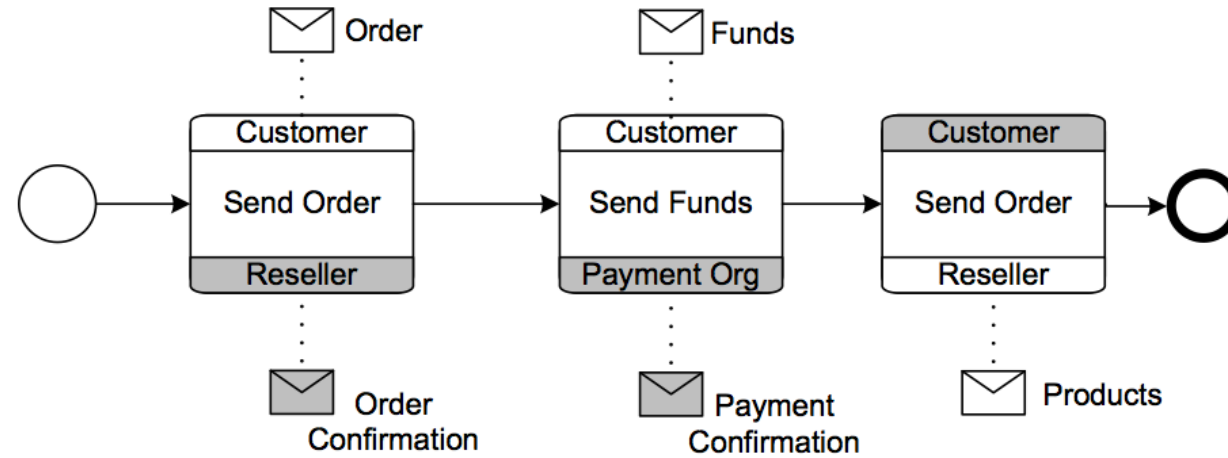


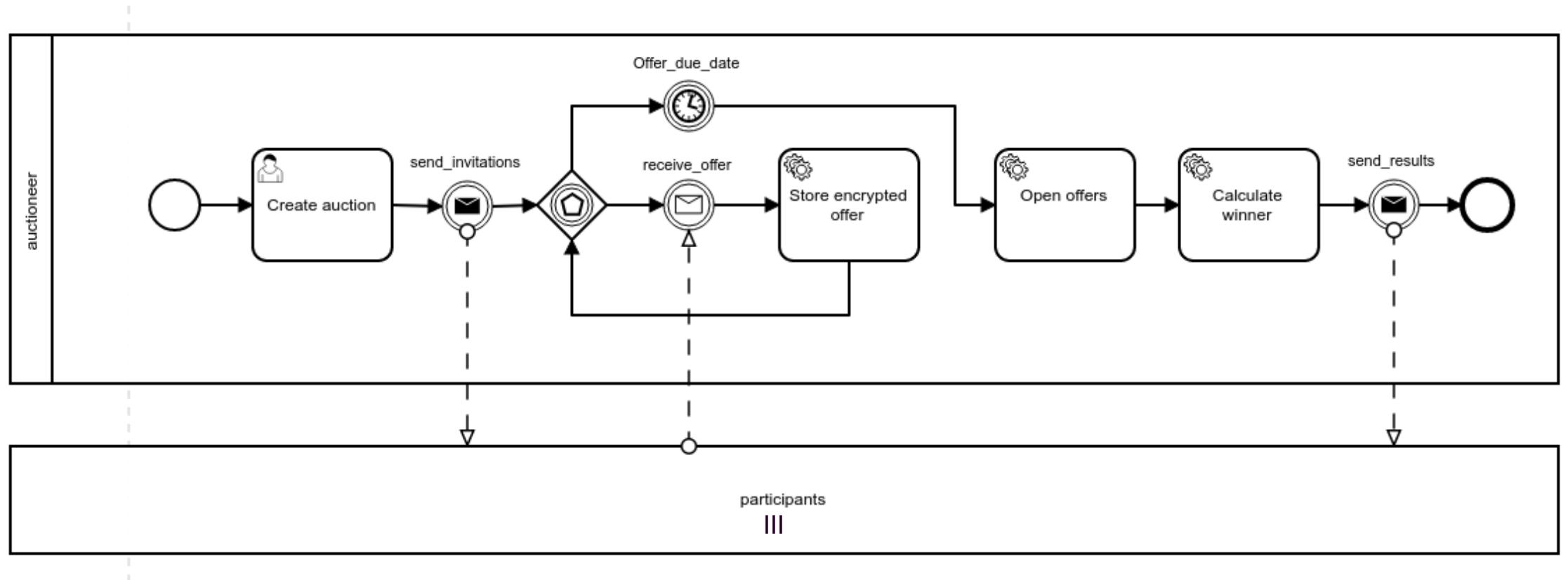
Fig. 5.42. Choreography diagram that is not enforceable, since the reseller cannot know if the customer has already send the funds

Assignment: BPMN auction scenario

- Create a BPMN diagram for an auction scenario involving bidders and the auction house. Model key interactions like bid submission, bid evaluation, winner determination, and payment processing.
- Form a group
- Time: 30 minutes
- Objective: Familiarize yourself with a complex BPMN topic involving multi pools
- Result: Present your BPMN diagram to the class
- Format: Create the diagram on paper or digitally
- Help: Ask the lecturer for assistance



Assignment: BPMN auction scenario example



Questions?



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